Hands-On 1

Alzheimer Data Preprocessing and KNN Classification

Goal: Build a leakage-safe preprocessing pipeline, train a K-nearest neighbors classifier, and use visual evidence to explain performance.

1. **Audit the raw data:**
   Load the CSV with pandas. Display the first five rows, shape, data types, missing counts, duplicate count, and unique values per column.

2. **Remove leakage and non-informative fields:**
   Inspect PatientID and DoctorInCharge, then remove both. Separate Diagnosis from the predictors.

3. **Inspect balance and unusual values:**
   Plot the Diagnosis counts with percentages. Create a 2 × 2 boxplot grid for BMI, SystolicBP, CholesterolTotal, and MMSE. Use the IQR rule to count potential outliers
   Do not delete outliers automatically. Comment on plausibility and class imbalance.

4. **Split and preprocess:**
   Use an 80/20 stratified split with random_state=42. Standardize the 15 features with more than 10 unique values, one-hot encode Ethnicity, and pass binary and ordinal fields through. Fit on X_train. Transform X_train and X_test. Confirm processed shapes are 1,719 × 35 and 430 × 35.

5. **Visualize the effect of scaling:**
   Make two scatterplots of MMSE versus CholesterolTotal: one
   in raw training units and one after training-only standardization. Color points by Diagnosis. In two sentences, explain what scaling changes and what it preserves.

6. **Train the KNN baseline:**
   Convert the transformed arrays to labeled DataFrames using get_feature_names_out(). Train KNeighborsClassifier(n_neighbors=5) and predict the untouched test set.

7. **Evaluate with plots:**
   Report accuracy, balanced accuracy, Alzheimer precision, Alzheimer recall, and macro F1. Create:
   - a confusion matrix labeled No Alzheimer and Alzheimer;
   - a 0-to-1 bar chart comparing all five metrics.

8. **Test k and explain one prediction:**
   Repeat training for k = 3, 5, 7, 9, 11, 15. Plot balanced accuracy and Alzheimer recall against k. Select one k and justify the trade-off.
   For one test patient, use kneighbors() to display the five distances and class votes. Optional: mark the query and its exact neighbors on a 2D PCA plot. PCA is for display only; neighbor search must use all processed features.

Concrete Preprocessing and Linear Regression

Goal: Build a leakage-safe linear regression workflow that predicts concrete compressive strength and uses residual evidence to
assess its limits

1. Load and clean the labels

Load the CSV with pandas. Display the first five rows and shape. Strip surrounding spaces from every column name, then rename the target to Concrete compressive strength (MPa).

2. Audit data quality

Report data types, missing counts, unique values, minimums, maximums, constant columns, and exact duplicate rows.

□ Do not impute when nothing is missing.

□ Do not delete the 25 duplicate flags without provenance showing they are errors.

3. Define predictors and target

Create X from the eight input columns and y from the cleaned MPa target. Confirm X is 1,030 x 8 and y contains 1,030 continuous values.

□ Explain why Age (day) remains quantitative even though it has only 14 unique values.

4. Explore distributions and relationships

Create a 30-bin histogram with KDE for the target and an annotated Pearson correlation heatmap for all variables.

- Describe skew, strong linear associations, and possible multicollinearity clues.
- State why correlation does not establish causation.

5. Split and fit the regression pipeline

Use an 80/20 split with random_state=42. Put StandardScaler and LinearRegression in one pipeline, fit on training data, and
predict the test set. (Confirm train and test shapes are 824 x 8 and 206 x 8.)

6. Compare with a mean baseline

Fit DummyRegressor(strategy='mean') on the training set. For both models, report MAE, RMSE, and R2.

7. Interpret standardized coefficients

Extract the fitted coefficients, sort them by absolute value, and draw a horizontal bar chart with a zero line. Use different
colors for positive and negative coefficients.

- Interpret one coefficient as predicted MPa change per one training-standard-deviation increase, holding other inputs fixed.

Treat coefficients as conditional model associations, not causal effects.

8. Diagnose prediction errors

Create a two-panel figure:

- actual versus predicted strength with a 45-degree reference line;
- residuals versus predicted strength with a zero-reference line.

Identify curvature, changing spread, clusters, or influential observations. Explain whether the straight-line additive model leaves systematic structure unexplained.